Formal Buddhist Logic

The catuṣkoṭi, non-perception, valid reasons, exclusion and the two truths, mechanized over PLN evidence

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October 15, 2026

Abstract

Indian Buddhist logicians drew distinctions, more than a millennium ago, that formal reasoning systems keep rediscovering under other names: evidence for a claim against evidence against it; "no evidence" against "evidence of absence"; denying a claim against asserting its opposite; a concept as what it excludes; conventional truth against ultimate truth. We give each of these distinctions an exact mathematical reading on carriers that an existing reasoning library already uses, chiefly the two-channel evidence of Probabilistic Logic Networks (PLN), and we prove the readings in Lean 4. The four corners of the catuṣkoṭi are exactly the values that double negation fixes on PLN evidence, and the logic of supported formulas over that evidence is First Degree Entailment. Dharmakīrti's rule of non-perception becomes one law with a single side condition, perceptibility, under which absence of evidence is evidence of absence; eight results already in the library turn out to be instances of it, among them Chalmers's "that's all" clause. The three marks of a valid reason, Dignāga's wheel of reasons, the apoha theory of meaning by exclusion, which meets the distinction calculus at the sorites, and Madhyamaka's two truths each receive a theorem with positive and negative controls. A final section separates thirteen kinds of "no value" that programming languages tend to collapse. Every theorem cited is proved without `sorry`, with no axioms beyond Lean's standard three.

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1 Introduction

The motivation is semantic clarity. Several techniques familiar from knowledge representation and automated reasoning, namely local closed-world assumptions, coverage-based scoring of candidate rules, paraconsistent treatment of conflicting data, and learning concepts from negative examples, are usually presented separately and justified case by case. The Buddhist logical tradition presents them as consequences of a small number of principles, stated with unusual precision about the conditions under which they hold. That precision is what a formal library can use: one law with an explicit side condition replaces several local rules with implicit ones.

We work over carriers that the Mettapedia library already uses for inference rather than over carriers invented for the purpose. The principal one is PLN's two-channel evidence `BinaryEvidence`, a pair (n^+, n^-) of extended non-negative reals recording evidence for and evidence against a proposition. Others are Kripke-style frames of information stages, concept lattices from a distinction calculus, and the theory–model Galois connection.

Each section follows the same pattern: the doctrine, quoted from a source; its formal reading; the theorem, stated mathematically; and the Lean statement, quoted from the library. The principal source is Stcherbatsky's *Buddhist Logic* [10], the classic account of the school of Dignāga and Dharmakīrti, read beside Kajiyama's translation of Mokṣākaragupta's *Tarkabhāṣā* [1] and, for Madhyamaka, the work of Garfield, Priest and Westerhoff [3, 2, 4]. Table 1 summarizes the correspondences.

Doctrine	Formal reading	Lean module
Catuṣkoṭi (four corners)	Double-negation-fixed values of PLN evidence; FDE as the logic of support	`DoubleNegationCorners`, `Tetralemma`
Two negations	Nelson's strong negation and intuitionistic negation on the truth order; the liar's two fixed points	`TwistNegation`, `Tetralemma`, `KripkeFixedPoint`
Non-perception	Absence equals non-perception exactly at perceptible stages; eight instances	`NonPerception`, `NonPerception.Instances`
Three marks; wheel of reasons	Soundness of the marks; induction as non-perception of a counterexample; the wheel as pervasion evidence	`ReasonValidity`, `DoubleNegationCorners`
Apoha	Concepts as closed classes of double exclusion; the dichotomy as a law of construction; learning by exclusion	`Exclusion`, `ConstructionMiddle`, `ExclusionLearning`
Two truths	Bubble/top layering of consequence; the theory–model Galois connection	`TwoTruths`, `Tetralemma`
Kinds of no value	Thirteen pairwise separated kinds; open and closed declarations; empty functions; parameterized nulls	`NoValueZoo`, `OpenDeclarations`, `EmptyFunctions`, `NullDesigns`

Table 1: Doctrines, their formal readings, and where the proofs live. Full module names are in Appendix A.

2 The four corners on evidence

2.1 The doctrine

Priest opens his study of the catuṣkoṭi as follows [2]:

> In early Buddhist logic, it was standard to assume that for any state of affairs there were four possibilities: that it held, that it did not, both, or neither. This is the catuskoti (or tetralemma).

He proposes First Degree Entailment (FDE) as its logic, on the strength of its four-valued semantics: "The four corners of truth and the Hasse diagram seem like a marriage made for each other in a Buddhist heaven."

2.2 Double negation reads off the corner

`BinaryEvidence` carries an information order, componentwise on (n^+, n^-), under which it is a frame. Its Heyting complement e^{c} saturates exactly the empty channels, and its double complement e^{cc} saturates every channel that carries any evidence.

Theorem 2.1 (Double negation is the four-corner readout)**.** *An evidence value is regular (fixed by double negation) exactly when each channel is* 0 *or* $\top$*. The regular values are the four saturated corners* $(0,0)$*,* $(\top,0)$*,* $(0,\top)$*,* $(\top,\top)$*, read as neither, true, false and both. For every* e*,* e^{cc} *is the corner of* e*. The regular elements form a four-element Boolean algebra, in bijection with the four corners and ordered by Belnap's knowledge order:* neither *below everything,* both *above everything,* true *and* false *incomparable.*

```
theorem isRegular_iff (e : BinaryEvidence) :
    Heyting.IsRegular e ↔ (e.pos = 0 ∨ e.pos = ⊤) ∧ (e.neg = 0 ∨ e.neg = ⊤)
theorem compl_compl_eq_saturatedCorner (e : BinaryEvidence) :
    e^cc = saturatedCorner (quadrant e)
noncomputable def regularEquivQuadrant : Heyting.Regular BinaryEvidence ≃ Quadrant
theorem saturatedCorner_le_iff (q q' : Quadrant) :
    saturatedCorner q ≤ saturatedCorner q' ↔ q = q' ∨ q = .neither ∨ q' = .both
```

In plain words, double negation forgets *how much* evidence there is and keeps *which sides* are touched. Classical reasoning about evidence is reasoning about which channels are touched.

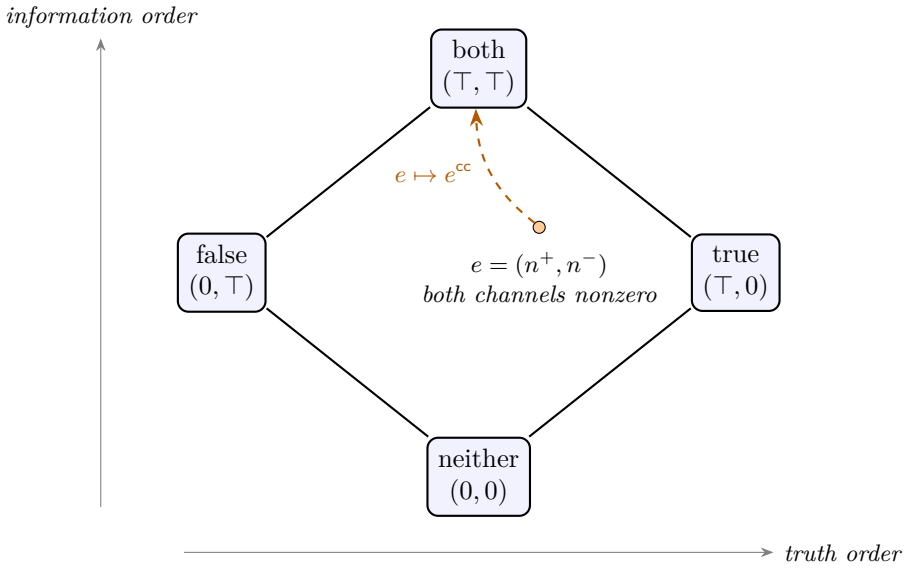


Figure 1: The four corners of the catuṣkoṭi as Belnap's four values [12] inside PLN evidence. Vertically, the information order; horizontally, the truth order. Double negation sends every evidence value to the corner whose channels it touches.

2.3 FDE is the logic of support

Say that a formula is *supported* by an evidence valuation when its positive channel is nonzero, and *refuted* when its negative channel is. Interpret conjunction and disjunction on the truth order and negation as the exchange of channels.

Theorem 2.2 (The catuṣkoṭi is FDE over evidence)**.** *For every valuation v of atoms in* `BinaryEvidence` *and every formula φ, support and refutation of φ are exactly Dunn's four-valued evaluation [13] of φ at the supports of the atoms. Hence the logic of supported formulas over PLN evidence is FDE, and magnitudes never affect satisfaction: they affect strength only.*

```
theorem support_refutation_evalEvidence (v : Atom → BinaryEvidence) :
    ∀ formula : AtomFormula Atom,
      ((evalEvidence v formula).pos ≠ 0 ↔ (atomSystem.eval (supportValuation v) formula).1) ∧
        ((evalEvidence v formula).neg ≠ 0 ↔ (atomSystem.eval (supportValuation v) formula).2)
```

Priest stipulates FDE for the catuṣkoṭi; here it is a theorem about a carrier the library already uses for inference: FDE is the support-level shadow of PLN evidence.

3 Two negations and the two tetralemmas

3.1 The doctrine

Westerhoff explains why Nāgārjuna's tetralemma, which rejects a statement, its negation, both and neither, is not simply inconsistent [4, p. 28]:

> Some are implicational negations (paryudāsapratiṣedha) which make an assertion about the object referred to ('the apple is not red' implies that it is some other colour), while others are non-implicational (prasajyapratiṣedha) and do not make such an assertion ('the force of gravity is not red' does not imply that it is some other colour).

3.2 Strong and intuitionistic negation

On the truth order of `BinaryEvidence` there are two negations. Strong negation $\sim a$, Nelson's constructible falsity [14], exchanges the channels: it asserts evidence against. Intuitionistic negation $\mathrm{inot}\, a$ asserts only that there is no support. These are the implicational and non-implicational negations respectively.

Theorem 3.1 (When absence of support counts against)**.** *Intuitionistic negation entails strong negation on the truth order exactly when the absence of support is full refutation.*

```
theorem inot_truthLE_cdNeg_iff (a : BinaryEvidence) :
    TruthLE (inot a) (∼a) ↔ (a.pos = 0 → a.neg = ⊤)
```

This is already the PLN form of the non-perception rule of Section 4: "no support" may be booked as evidence against only where the source is complete.

3.3 The two extremes of the theory lattice

Theorem 3.2 (Positive and negative tetralemma)**.** *Over evidence valuations:*

1. *the theory of all formulas has as models exactly the valuations that give every atom both support and refutation; there all four corners of every claim are supported;*
2. *exactly one valuation satisfies nothing, the one that is* neither *at every atom; there all four corners of every claim are unsupported, and all four are denied by intuitionistic negation;*
3. *among consistent valuations, the theory of all formulas has no model.*

```
theorem models_univ_eq_both :
    models (evidenceSat (Atom := Atom)) univ =
      {v | ∀ atom, (v atom).pos ≠ 0 ∧ (v atom).neg ≠ 0}
theorem corners_supported_of_degenerate {v : Atom → BinaryEvidence}
    (degenerate : Degenerate (evidenceSat (Atom := Atom)) v) (formula : AtomFormula Atom) :
    ∀ corner ∈ corners (evalEvidence v formula), Supported corner
theorem theoryOf_singleton_eq_empty_iff (v : Atom → BinaryEvidence) :
    theoryOf (evidenceSat (Atom := Atom)) {v} = ∅ ↔ v = neitherValuation
theorem corners_unsupported_at_neither (formula : AtomFormula Atom) :
    ∀ corner ∈ corners (evalEvidence (Atom := Atom) neitherValuation formula),
      ¬ Supported corner
theorem corners_denied_at_neither (formula : AtomFormula Atom) :
    ∀ corner ∈ corners (evalEvidence (Atom := Atom) neitherValuation formula),
      Supported (inot corner)
theorem modelsIn_consistent_univ_eq_empty [Nonempty Atom] :
    modelsIn (evidenceSat (Atom := Atom)) consistentValuations univ = ∅
```

Priest introduces a fifth value for "none of the four", because *neither* is itself one of the four [6]; Kapsner questions whether that value can be one more truth value of the same kind [7]. On evidence the difficulty splits by negation: with the corners stated using the presupposing (intuitionistic) negation, all four are denied at the *neither* valuation, consistently, without a fifth object-level value. What a fifth value genuinely captures, an evaluation with no result at all, is a different kind of "nothing" and belongs with the kinds of Section 8.

3.4 The liar's two escapes

Garfield and Priest read Nāgārjuna's limit contradictions dialetheically [3]; others read them as presupposition failures. In Kripke's theory of truth [8] these are the two extreme fixed points of one monotone jump; when the base facts are bivalent, conflation [15] exchanges them.

Theorem 3.3 (Paracomplete and dialetheic readings are dual)**.** *A judgment is undetermined in the least fixed point exactly when it both holds and fails in the greatest; the liar is* both *in the greatest fixed point.*

```
theorem undetermined_least_iff_both_greatest (judgment : Judgment) :
    (judgment ∉ system.least.holds ∧ judgment ∉ system.least.fails) ↔
      (judgment ∈ system.greatest.holds ∧ judgment ∈ system.greatest.fails)
theorem liar_both_greatest {liar : Judgment} (isLiar : system.spec liar = .neg (.truth liar)) :
    liar ∈ system.greatest.holds ∧ liar ∈ system.greatest.fails
```

The choice between the two readings is a choice of fixed point, not of foundations.

4 Non-perception

4.1 The doctrine

The *Tarkabhāṣā* of Mokṣākaragupta, in Kajiyama's translation [1], illustrates the logical mark of non-cognition (§13):

> In this place there is no jar, because it is not cognized though it is by nature perceptible.

The qualifier carries the whole weight (§13.3):

> . . . the mere non-occurrence of cognition regarding objects which are inaccessible in space, time and essence, does not establish practical activities referring to the negation [of the object concerned]; such objects are illustrated by Mt. Sumeru [which is spatially inaccessible], the future emperor Śaṅkha [who is inaccessible in time] and a ghost [which is inaccessible in essence].

Kajiyama's note summarizes Dharmakīrti's two kinds of non-cognition: "the former leads to the result through the absence of the proof, while the latter through the presence of the counter-proof; i.e. the former through doubt while the latter through the definite knowledge of absence." And (§13.4): "With regard to a future experience, however, non-cognition is itself doubtful and accordingly cannot be a means of valid knowledge."

Stcherbatsky explains what "perceptible" means for an absent thing, quoting Dharmottara [10, p. 365]:

> The perception of an absent thing is impossible, it is a contradiction. If it is perception, the thing is present, it cannot be absent. But how is it present? It is present in imagination and that means that all the conditions necessary for its perception are fulfilled. It would be necessarily perceived if it were present; but it is absent and therefore it is only imagined, but not perceived, it is perceived in imagination. [...] «How is it possible for an object, say a jar, to be perceptible, when it is absent? It is said to be perceptible, although it is absent, because its perceptibility is imagined! We imagine it in the following way: "If it were present on this spot, it certainly would have been perceived".»

4.2 The law

Definition 4.1. A *frame* is a set of stages W with a reflexive relation Future and a predicate Perceived. At a stage w:

- the thing is *absent* if no future stage perceives it;
- the stage is *perceptible* if whatever a future stage perceives is already perceived at w;
- the stage is *in doubt* if the thing is neither perceived nor absent.

Perceptibility is Dharmottara's counterfactual made exact: "if it were present on this spot, it certainly would have been perceived" becomes "whatever a later stage would perceive is perceived already".

```
structure Frame (W : Type u) where
  Future : W → W → Prop
  future_refl : ∀ w, Future w w
  Perceived : W → Prop
def Absent (w : W) : Prop := ∀ w', F.Future w w' → ¬ F.Perceived w'
def Perceptible (w : W) : Prop := ∀ w', F.Future w w' → F.Perceived w' → F.Perceived w
def Doubt (w : W) : Prop := ¬ F.Perceived w ∧ ¬ F.Absent w
```

Theorem 4.2 (Non-perception)**.** *At a perceptible stage, absence is exactly non-perception. Perceptibility is exactly the condition under which non-perception establishes absence. A stage is in doubt exactly when it is not perceptible, and every stage is perceived, absent or in doubt.*

```
theorem absent_iff_not_perceived {w : W} (perceptible : F.Perceptible w) :
    F.Absent w ↔ ¬ F.Perceived w
theorem perceptible_iff {w : W} : F.Perceptible w ↔ (¬ F.Perceived w → F.Absent w)
theorem doubt_iff_not_perceptible {w : W} : F.Doubt w ↔ ¬ F.Perceptible w
theorem perceived_or_absent_or_doubt (w : W) :
    F.Perceived w ∨ F.Absent w ∨ F.Doubt w
```

4.3 Eight instances, and Chalmers's "that's all"

The same law appears in the library about eight times under other names, each with its own reflection or completeness condition playing the role of perceptibility:

- absence in presheaf models of events;
- permitted absence of reactions in a scoped process calculus;
- refutation by a decision backend, which is allowed only when the backend is closed;

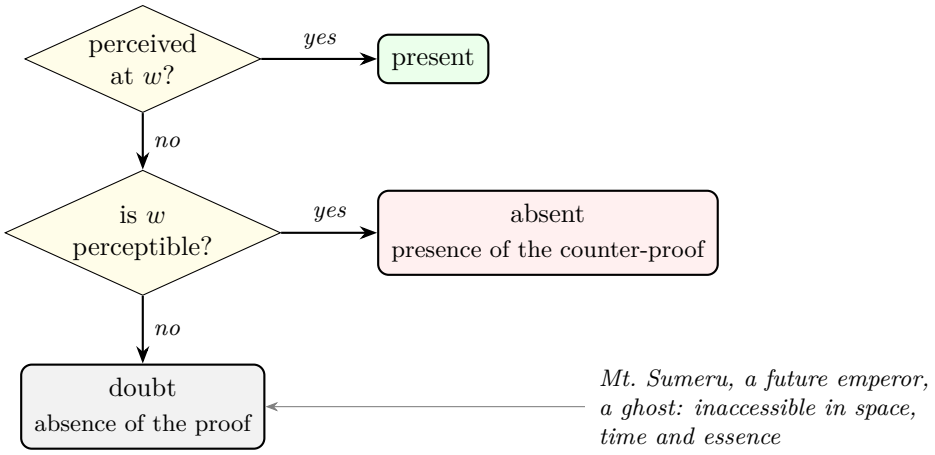


Figure 2: The non-perception law. Not perceiving establishes absence only at a perceptible stage; elsewhere it leaves doubt.

- answers outside a decidable fragment;
- fuel-bounded runs, where exhaustion is doubt and not refusal;
- underfunded assays, which stay quiet in both directions;
- scenario bases, through Chalmers's "that's all" clause;
- concept construction at a vantage.

Each is re-derived from the law as a corollary. Two of them:

```
theorem thatsAll_iff_perceptible (actual : S) :
    I.ThatsAll positive actual ↔
      ∀ sentence, (scenarioFrame I positive sentence).Perceptible actual
theorem exhaustedAsRefusal_books_doubt {logical : Query → Option Answer}
    {run : Query → Nat → RunOutcome Answer} {query : Query} {fuel : Nat} {answer : Answer}
    (exhausted : run query fuel = .exhausted) (answerable : logical query = some answer) :
    (runFrame logical run query fuel).Doubt false ∧
      exhaustedAsRefusal (run query fuel) = some none
```

The first is worth reading slowly. In Chalmers's scrutability base PQTI [9], the "that's all" truth T says that the world is no more than the base truths require. The theorem says that T holds at the actual scenario exactly when every sentence is perceptible there. So T is perceptibility made global: with it, what the base does not contain is absent, and the world is closed; without it, the frontier beyond the base stays in doubt, and the world is open. The closed world and the open-ended world meet at exactly one clause.

4.4 A rule for evidence

The PLN consequence is a single bookkeeping rule. Absence of support may be booked as negative evidence only for a proposition the source is complete for; otherwise it stays in the *neither* corner $(0, 0)$. A certificate of exhaustion is itself positive evidence ("the presence of the counter-proof"), which is why a completed search can refute while a timed-out one cannot.

5 Valid reasons

5.1 The three marks

The *Tarkabhāṣā* (§10) states the triple characteristic of a logical mark h for the property s of a subject p [1]:

> 1) Its [h's] definite (niścitam) presence in all [the members of the class of] the locus (p) of inference [as well as in other classes].
> 2) Its definite presence only in things similar to p [sapakṣa i.e. the members of the class of s].
> 3) Its definite, absolute absence in the anti-pakṣa [vipakṣa i.e. any member of the class incompatible with s].

The second is named *anvaya* (positive pervasion), the third *vyatireka* (negative pervasion).

Theorem 5.1 (Soundness of the marks)**.** *If the reason holds of the subject and is universally absent wherever the property is absent, the property holds of the subject.*

```
theorem sound_of_marks {subject : α} {reason property : α → Prop} (paksha : reason subject)
    (vyatireka : ∀ x, ¬ property x → ¬ reason x) : property subject
```

Stcherbatsky adds a remark that the wheel below makes exact [10, p. 244]:

> It is easily seen that the second and the third rule mutually imply each the other. If the reason is present in the similar instances only, it also is absent from every dissimilar case. And if it is absent from every dissimilar case, it can be present in similar instances only, although not necessarily in all of them. Nevertheless both rules must be mentioned, because, although in a correct inference the application of the one means the application of the other, in a logical fallacy their infringements carry sometimes different results.

The difficulty is the word "absolute": observed absence on examined cases is not universal absence. The library makes the step from one to the other an instance of non-perception.

Theorem 5.2 (Induction is non-perception of a counterexample)**.** *Form a two-stage frame whose stages are the examined instances and all instances, and which perceives a counterexample when one is found. The examination is perceptible exactly when every counterexample would have been examined; then observed exclusion is universal. Otherwise an unexamined counterexample leaves the inference in doubt.*

```
theorem inductionFrame_perceptible_iff :
    (inductionFrame examined reason property).Perceptible false ↔
      ((∃ x, reason x ∧ ¬ property x) → ∃ x ∈ examined, reason x ∧ ¬ property x)
theorem universal_of_observed
    (perceptible : (inductionFrame examined reason property).Perceptible false)
    (observed : ∀ x ∈ examined, ¬ property x → ¬ reason x) :
    ∀ x, ¬ property x → ¬ reason x
theorem doubt_of_unexamined_counterexample
    (observed : ∀ x ∈ examined, ¬ property x → ¬ reason x)
    {c : α} (carries : reason c) (lacks : ¬ property c) :
    (inductionFrame examined reason property).Doubt false
```

5.2 Dignāga's wheel of reasons

Dignāga tabulated the possible positions of a reason by its presence among similar instances (those with the property) and among dissimilar instances (those without it), each "in all", "in some" or "in none". Stcherbatsky describes the table [10, p. 321]:

> All the possible positions of the reason between similar and dissimilar instances are computed in this table, according to a mathematical principle. The result is that there are only nine positions of the Reason, neither more, nor less. Of them two only represent right reasons, the remaining seven are fallacies. Out of these seven, two represent the fallacy at its maximum, they are the contradictorily opposed part of right reason, and are called «contrary» or «inverted» reasons. The five remaining ones are «uncertain», because the position of the middle term between similar and dissimilar instances is not definite; [. . .]

Figure 3 gives the table in the form the library uses. Of the five uncertain positions, the library names the two extreme ones separately, as *too wide* (present everywhere) and *too narrow* (present nowhere).

presence among similar instances \ *presence among dissimilar instances*	*none*	*some*	*all*
all	right	uncertain	too wide
some	right	uncertain	uncertain
none	too narrow	inverted	inverted

Figure 3: Dignāga's wheel as defined in the library: two right cells, two inverted, one too wide, one too narrow and three uncertain. Reading presence among similar instances as support and presence among dissimilar instances as refutation, *right* is the *true* corner, *inverted* the *false* corner, *too narrow* the *neither* corner, and *too wide* and *uncertain* are the *both* corner. The four cells built from "all" and "none" are the saturated corners of Figure 1; "some" is a magnitude that double negation forgets.

```
def wheel : Occurrence → Occurrence → WheelVerdict
  | .all, .all => .tooWide
  | .none, .all => .inverted
  | .some, .all => .uncertain
  | .all, .none => .right
  | .none, .none => .tooNarrow
  | .some, .none => .right
  | .all, .some => .uncertain
  | .none, .some => .inverted
  | .some, .some => .uncertain
theorem wheel_quadrant (similar dissimilar : Occurrence) :
    (wheel similar dissimilar).quadrant = supportQuadrant similar.present dissimilar.present
```

Theorem 5.3 (The wheel is pervasion evidence)**.** *Count the reason among the similar and among the dissimilar instances; this is the evidence for the implication from reason to property. A reason is*

in a right *position exactly when that evidence has strength* 1*, and* too narrow *exactly when there is no evidence at all. For every nonzero prior, finite counts have confidence strictly below* 1*.*

```
theorem wheelOf_right_iff_strength :
    wheelOf similar dissimilar reason = .right ↔
      (pervasionEvidence similar dissimilar reason).toStrength = 1
theorem wheelOf_tooNarrow_iff_total :
    wheelOf similar dissimilar reason = .tooNarrow ↔
      (pervasionEvidence similar dissimilar reason).total = 0
theorem pervasion_confidence_lt_one {κ : ENNReal} (prior : κ ≠ 0) :
    BinaryEvidence.toConfidence κ (pervasionEvidence similar dissimilar reason) < 1
```

Counting never certifies a pervasion. Nor does a right cell settle the subject, which belongs to neither class: three instances already give a reason in a right cell whose subject carries the reason and lacks the property (`right_cell_not_conclusive`). In the language of rule learning, a *too narrow* reason, present only in the subject, is a rule that covers nothing but its own training example: it carries no evidence at all.

5.3 The three kinds of reason

Dharmakīrti's *Nyāyabindu* (II.19, quoted in the *Tarkabhāṣā*) divides reasons into three kinds: "Among [the three kinds of logical marks] the two [i.e. the identical and causal marks] are for establishing the existence of real entities, the other one [i.e. the mark of non-cognition] is the probans for negation." Each supplies the certificate of pervasion that counting cannot.

The *identity* (essence) mark is a reason whose concept entails the property's, so the pervasion holds in every structure. Over evidence, a conjunction is an essence reason for each of its conjuncts, and a disjunction is not one for its disjuncts.

```
theorem essence_pervasion {reasonSentence propertySentence : Sent}
    (entails : Entails Sat {reasonSentence} propertySentence) (m : Str)
    (carries : Sat m reasonSentence) : Sat m propertySentence
theorem essence_conj (first second : AtomFormula Atom) :
    Entails evidenceSat {.conj first second} first
theorem not_essence_disj :
    ¬ Entails (evidenceSat (Atom := Bool)) {.disj (.truth true) (.truth false)} (.truth true)
```

The mark of non-cognition is the law of Section 4, applied to the counterexample as above. The *causal* mark is proved over structural causal models, with a control showing exactly when it fails.

```
theorem effect_pervasion {mechanisms : Mechanisms Key Bool} {effect cause : Key}
    (requires : RequiresCause mechanisms effect cause) {store : Key → Bool}
    (solution : IsSolution mechanisms store) (seen : store effect = true) :
    store cause = true
theorem smokeMachine_breaks_pervasion :
    ¬ RequiresCause (surgery smokeMachine (mechanisms false)) .smoke .fire ∧
      machineSmoke .smoke = true ∧ machineSmoke .fire = false
```

If the mechanism of smoke produces it only where there is fire, every solution with smoke has fire. Under the intervention that installs a smoke machine, smoke without fire is a solution: the effect reason fails exactly when its mechanism is replaced.

6 Meaning by exclusion

6.1 The doctrine

Katsura summarizes Dignāga's theory of meaning [5]:

> In the Fifth Chapter of the P(ramāṇa)S(amuccaya) Dignāga presents the theory of anyāpoha (exclusion of others) which may be regarded as his most original contribution to Indian logic. Dignāga declares in the introductory verse (K. 1) that verbal cognition (śabda) is nothing but inference (anumāna) because both of them function on the principle of anyāpoha.

Stcherbatsky states the thesis in its starkest form, reporting Śāntarakṣita and Kamalaśīla [10, p. 471]:

> [. . .] the words express their own meaning through negation. They are therefore negative. Without negation they express nothing, they can express something only dialectically, i.e. in couples of mutual negation.

6.2 Concepts as double exclusion

Apoha relies on a dichotomy that Stcherbatsky's account of Dharmakīrti states as the law of contradiction [10, pp. 403 and 417]:

> «There is contradiction, says he, in a couple whose essence is posited in a complete mutual exclusion, as, e.g., existence and non-existence». Complete mutual exclusion means mutual exclusion without anything intermediate.
>
> When we cognize a patch of blue in the judgment «this is blue», we then, owing to a Primordial Function of Productive Imagination, construct out of the Universe of Discourse two parts, the blue and the non-blue. Everything that is not referred to the blue will be necessarily in the non-blue. There can be no third possibility, nothing in the middle.

For an observer, exclusion is a symmetric irreflexive relation of witnessed apartness. The exclusion of a class is everything apart from all of its members, and a concept is a class equal to its double exclusion. These closed classes are the propositions of Goldblatt's semantics for orthologic [11].

Theorem 6.1 (Apoha)**.** *Treble exclusion is exclusion. Concepts by exclusion are exactly the extents of the exclusion context. Nothing lies in a class and in its exclusion, and a class together with its exclusion generates every instance. Yet every instance lies in each concept or in its exclusion exactly when non-exclusion is transitive.*

```
theorem excluded_excluded_excluded (S : Set V) :
    excluded excludes (excluded excludes (excluded excludes S)) = excluded excludes S
theorem isConcept_iff_isExtent (S : Set V) : IsConcept excludes S ↔ Order.IsExtent excludes S
theorem inter_excluded_eq_empty (S : Set V) : S ∩ excluded excludes S = ∅
theorem exclusionClosure_union_excluded (S : Set V) :
    excluded excludes (excluded excludes (S ∪ excluded excludes S)) = univ
theorem union_excluded_eq_univ_iff :
    (∀ S, IsConcept excludes S → S ∪ excluded excludes S = univ) ↔
      ∀ x y z, ¬ excludes x y → ¬ excludes y z → ¬ excludes x z
```

```
-- control: three instances, the middle one indistinguishable from both ends
theorem middle_in_neither :
    (1 : Fin 3) ∉ ({0} : Set (Fin 3)) ∪ excluded tolerance.Apart {0}
```

The theorem separates the two halves of the quoted law. "Nothing in the middle" is a law of construction: a concept and its exclusion together generate the whole universe of discourse. It is not a law of particulars: a particular may fall between a concept and its exclusion whenever indistinguishability is not transitive, as the middle of three instances does when it is indistinguishable from both ends while the ends are apart. The blue and the non-blue exhaust what can be constructed, not what can be seen. Exclusion is not resemblance.

6.3 Perception, construction and the sorites

The distinction calculus [16] takes distinction as primitive: an observer is a graded similarity on its instances, reflexive and symmetric. Its author states where the structure lies:

> [...] indistinguishability is reflexive and symmetric but not transitive, and the failure of transitivity, the sorites, is where all the nontrivial structure lives.

The previous theorem says the same from the other side: particulars fall between a concept and its exclusion exactly when indistinguishability is not transitive. The two meet in the closure of an observer, which identifies whatever a chain of indistinguishable steps connects. Read the observer as perception and its closure as conceptual construction. Dharmakīrti's dichotomy is then a law of the closure, and it fails for the observer exactly on the sorites.

Theorem 6.2 (Construction removes the middle)**.** *Let exclusion be symmetric and irreflexive, and let the* constructed *exclusion separate two instances when no chain of indistinguishable steps connects them.*

1. *A class and its exclusion cover everything exactly when the class is closed under indistinguishability.*
2. *The concepts of the observer that have no middle are exactly the concepts of its construction.*
3. *An instance in the middle of a concept is indistinguishable from a member and from an excluded instance, which exclude each other: every middle is the middle of a sorites triple.*

```
def constructed (excludes : V → V → Prop) (x y : V) : Prop :=
  ¬ Relation.EqvGen (fun a b => ¬ excludes a b) x y
theorem union_excluded_eq_univ_iff_saturated (S : Set V) :
    S ∪ excluded excludes S = univ ↔ Saturated excludes S
theorem noMiddle_concept_iff (irrefl : ∀ x, ¬ excludes x x) (S : Set V) :
    (IsConcept excludes S ∧ S ∪ excluded excludes S = univ) ↔
      IsConcept (constructed excludes) S
theorem middle_sorites {S : Set V} (concept : IsConcept excludes S) {y : V}
    (notIn : y ∉ S) (notOut : y ∉ excluded excludes S) :
    ∃ x ∈ S, ∃ z ∈ excluded excludes S, ¬ excludes x y ∧ ¬ excludes y z ∧ excludes x z
```

Graded observers sharpen the picture. Read a similarity at a level s, counting two instances apart when their similarity is below s. The distinction calculus has two transitivity classes: *metric*, where distances obey the triangle inequality, and *ultrametric*, where similarity across two steps is at least that of the weaker step.

Theorem 6.3 (The dichotomy at every resolution)**.** *Every concept at every level $s \in (0, 1]$ has no middle exactly when the observer is ultrametric. The metric law is not enough: a metric observer on three instances, with neighbours at similarity $3/4$ and the ends at $1/2$, has a concept with a middle at level $3/4$.*

```
theorem noMiddle_all_levels_iff_ultrametric (a : Tolerance W R) :
    (∀ s : R, 0 < s → s ≤ 1 → ∀ S, IsConcept (a.LevelApart s) S →
        S ∪ excluded (a.LevelApart s) S = univ) ↔ a.Ultrametric
theorem metricTolerance_metric : metricTolerance.Metric
theorem metricTolerance_not_ultrametric : ¬ metricTolerance.Ultrametric
theorem metric_middle :
    IsConcept (metricTolerance.LevelApart (3 / 4)) {0} ∧
      (1 : Fin 3) ∉ ({0} : Set (Fin 3)) ∪ excluded (metricTolerance.LevelApart (3 / 4)) {0}
```

In the three-instance observer of the previous control, construction connects the two ends through the middle and separates nothing; the concept $\{0\}$, which has a middle for the observer, is not a concept of the construction. Construction buys the dichotomy by forgetting the distinction between the ends.

6.4 Learning by exclusion

Learning from examples then takes the familiar form of version spaces [17], with concepts by exclusion as the hypotheses.

Theorem 6.4 (Version spaces of concepts)**.** *A concept consistent with positive and negative examples exists exactly when every positive example excludes every negative one. The consistent concepts are exactly those between the concept generated by the positives and the exclusion of the negatives.*

```
theorem consistent_iff {pos neg S : Set V} :
    Consistent excludes pos neg S ↔
      IsConcept excludes S ∧ excluded excludes (excluded excludes pos) ⊆ S ∧
        S ⊆ excluded excludes neg
theorem exists_consistent_iff {pos neg : Set V} :
    (∃ S, Consistent excludes pos neg S) ↔ Separated excludes pos neg
```

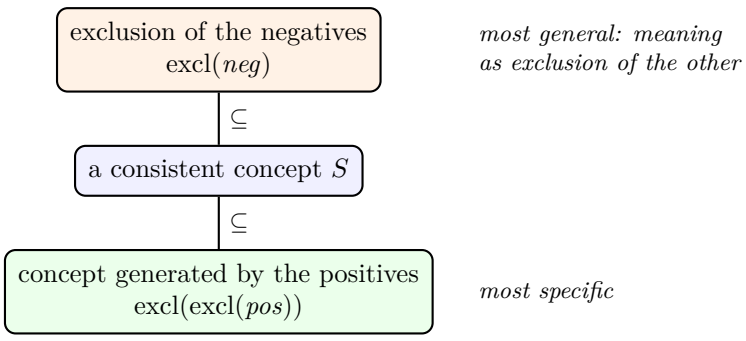


Figure 4: The consistent hypotheses of learning by exclusion form an interval of concepts.

A small example shows the interval. On a circle of six hues where each hue is indistinguishable from its two neighbours, the observer has no metric law: 0 and 2 are apart, though each is indistinguishable from 1. The concept of 0 is $\{0\}$, while its resemblance class is $\{0, 1, 5\}$. From the positive example 0 and the negative example 3, the consistent hypotheses are exactly four, from the concept of 0 up to its resemblance class:

```
-- the six-hue circle
theorem conceptOf_zero : conceptOf hues.Apart 0 = {0}
theorem resemblance_zero : {y | ¬ hues.Apart 0 y} = {0, 1, 5}
theorem consistent_iff (S : Set (Fin 6)) :
    Consistent hues.Apart {0} {3} S ↔ S = {0} ∨ S = {0, 1} ∨ S = {0, 5} ∨ S = {0, 1, 5}
```

An experiment over random observers illustrates the point in numbers (Table 2). Each row draws 200 observers on n instances, in which two distinct instances are indistinguishable with probability p, and asks how concepts compare with resemblance. For learning, each observer contributes one target concept with a middle; half of its members, chosen at random, are the positive examples, and half of a pool, chosen at random, are the negative examples. The pool is either the exclusion of the target or its complement.

			negatives from the exclusion				negatives from the complement			
n, p	concept $\subsetneq$	middle	cons.	spec.	gen.	size	cons.	spec.	gen.	size
8, 0.3	86.6%	92.6%	100%	8%	31%	5.4	56.5%	7%	29%	3.1
8, 0.5	98.1%	99.0%	100%	18%	46%	3.4	28.5%	5%	21%	1.6
10, 0.3	94.4%	98.0%	100%	9%	19%	7.1	34.5%	4%	23%	3.1
10, 0.5	99.6%	99.9%	100%	13%	35%	4.3	15.0%	7%	27%	1.7

Table 2: Learning by exclusion on random observers. *concept* $\subsetneq$: the concept of an instance is strictly inside its resemblance class. *middle*: a proper concept has instances in neither it nor its exclusion. *cons.*: a consistent hypothesis exists. Among the cases where one exists, *spec.* and *gen.*: the most specific or the most general hypothesis is the target; *size*: the mean number of consistent hypotheses.

Exclusion is almost never resemblance, and almost every concept has a middle. Negative examples taken from what a concept excludes always admit a consistent hypothesis, as the theorem requires; negative examples taken from mere non-membership fail to in 43–85% of cases. "Not a cow" must be learned as "excluded by cow", not as "not observed to be a cow". Where a hypothesis exists, the most general one, meaning as exclusion of the other, recovers the target between two and six times as often as the most specific.

7 The two truths

7.1 The doctrine

Garfield and Priest present Nāgārjuna's doctrine [3]:

> Central to Nagarjuna's view is his doctrine of the two realities. There is, according to Nagarjuna, conventional reality and ultimate reality. Correspondingly, there are two truths: conventional truth, that is, the truth about conventional reality, and ultimate truth, or the truth about the ultimate reality—qua ultimate reality.

They cite the verse that forbids taking emptiness itself as a view (*Mūlamadhyamakakārikā* XIII.8, Garfield's translation):

> The Victorious ones have said
> That Emptiness is the relinquishing of all views.
> For whomever emptiness becomes a view
> That one will accomplish nothing.

7.2 Madhyamaka: conventions carve the ultimate

Fix a universe U of structures and a theory T. A *convention* Φ carves U to the structures satisfying Φ. The *conventional* truths under Φ are the consequences of T in the carved universe, and the *ultimate* truths are the consequences of T in U itself.

Theorem 7.1 (Two truths as layering)**.** *Every ultimate truth holds under every convention, and a conventional truth is an ultimate truth of the enlarged theory: a convention is a commitment, not a fact. When conventions cover the universe, the ultimate is their common part. Two disagreeing conventions, each with a model, make neither of their claims ultimate.*

```
theorem ultimate_subset_conventional (U : Set Str) (Φ T : Set Sent) :
    ultimate Sat U T ⊆ conventional Sat U Φ T
theorem conventional_eq_ultimate (U : Set Str) (Φ T : Set Sent) :
    conventional Sat U Φ T = ultimate Sat U (Φ ∪ T)
theorem ultimate_eq_iInter_of_covers {ι : Sort uι} {U : Set Str} {conventions : ι → Set Sent}
    (covers : Covers Sat U conventions) (T : Set Sent) :
    ultimate Sat U T = ⋂ i, conventional Sat U (conventions i) T
theorem not_mem_ultimate_of_disagree {U : Set Str} {Φ Ψ T : Set Sent} {φ ψ : Sent}
    (holdsΦ : φ ∈ conventional Sat U Φ T) (holdsΨ : ψ ∈ conventional Sat U Ψ T)
    (hostedΦ : (modelsIn Sat (carve Sat U Φ) T).Nonempty)
    (hostedΨ : (modelsIn Sat (carve Sat U Ψ) T).Nonempty)
    (incompatible : ∀ m ∈ U, Sat m φ → Sat m ψ → False) :
    φ ∉ ultimate Sat U T ∧ ψ ∉ ultimate Sat U T
```

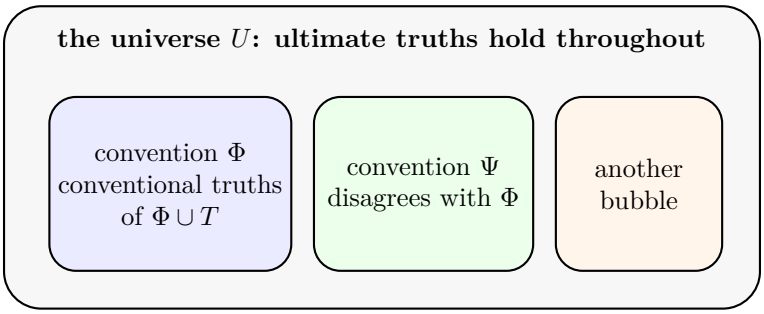


Figure 5: Madhyamaka's two truths as the bubble/top layering. Conventions carve bubbles in the universe; what holds throughout the universe holds in every bubble, while a bubble's truths lift only as commitments.

This is the layering a type theory with local definitional equalities needs: each bubble is a convention carving the ambient universe, what the top establishes holds in every bubble, and a bubble's equalities never silently become global. The empty convention carves nothing away, so its conventional truths are exactly the ultimate ones.

Over evidence the extremes of Section 3.3 complete the picture. The theory of all formulas has no consistent model: no consistent standpoint holds every view. At the other extreme exactly one valuation holds no view at all, the one that is *neither* at every atom, and it satisfies no nonempty convention. So a family of conventions covers all evidence only if one of them is the empty convention:

```
theorem conventional_empty (U : Set Str) (T : Set Sent) :
    conventional Sat U ∅ T = ultimate Sat U T
theorem exists_empty_of_covers_univ {ι : Sort*} {conventions : ι → Set (AtomFormula Atom)}
    (covers : Covers evidenceSat univ conventions) : ∃ i, conventions i = ∅
```

Read against *Mūlamadhyamakakārikā* XIII.8, the standpoint that relinquishes all views is reached by no view; it is a point of the space of standpoints, not one more theory in it. The two truths need it: without the empty convention, the conventions never exhaust what evidence can be.

7.3 Dharmakīrti: particulars and universals

Dharmakīrti's two realities are a different structure. Stcherbatsky's section "Two realities" opens with the definition [10, p. 69]:

> Existence, real existence, ultimate existence is nothing but efficiency. Whatsoever is causally efficient is real. The non-efficient is unreal, it is a fiction. [...] Only the present, the «here», the «now», the «this» are real.

The ultimately real is the particular, a structure; the conventionally real is the universal, a class of structures. This is the theory–model Galois connection, and over evidence it has a sharp consequence: two valuations that satisfy exactly the same formulas, so that no concept separates them, can still differ in strength.

```
theorem no_concept_separates {K : Set (Atom → BinaryEvidence)}
    (extent : Order.IsExtent evidenceSat K) (member : leaning ∈ K) : doubting ∈ K
theorem strength_leaning_ne_doubting (atom : Atom) :
    (leaning atom).toStrength ≠ (doubting atom).toStrength
```

Here `leaning` gives every atom evidence $(2, 1)$ and `doubting` gives $(1, 2)$: the same corner, strengths $2/3$ and $1/3$. What evidence does in inference, its efficacy, Dharmakīrti's mark of the real, is finer than any concept of it.

8 Kinds of "no value"

A recurring source of trouble in programming languages is that distinct kinds of "no value" are collapsed into one: a missing result, an error, an unevaluated call, an explicit `None`, an undetermined judgment, a refutation, an empty type. The precision about absence that the previous sections formalize applies directly.

8.1 Thirteen kinds, kept apart

The library lists thirteen kinds, each with a witness on its own carrier, and proves that every two kinds are separated by an observation.

```
inductive Kind where
  | zeroAnswers | unitDatum | emptyCollection | errorValue | inertCall | noneValue
  | emptyDatum | neither | undetermined | incomplete | refuted
  | closedEmptyType | openDeclaredType
theorem Kind.separated {k k' : Kind} (different : k ≠ k') :
    ∃ observe : Witness → Prop, observe k.witness ∧ ¬ observe k'.witness
```

One collapse is instructive because it is also a cost. An evaluator that erases error alternatives whenever a sibling alternative succeeds identifies kinds the list keeps apart, and the rule is not monotone in the alternatives, so no producer that emits results as they arrive can compute it: every level of evaluation must buffer.

```
theorem heNormalize_not_monotone :
    ¬ ∀ bag bag' : Multiset Pattern, bag ≤ bag' → heNormalize bag ≤ heNormalize bag'
theorem no_streaming_heNormalize (emit : Multiset Pattern → Multiset Pattern)
```

```
    (monotone : ∀ bag bag', bag ≤ bag' → emit bag ≤ emit bag') :
    ¬ ∀ bag, emit bag = heNormalize bag
```

8.2 Declared without constructors

A type declared without constructors is not an empty type. Its theory is the theory of an arbitrary set, possibly empty; adding a constructor later is conservative exactly when one already exists; and only *closing* the declaration, stating that these are all the constructors, makes non-membership of other names refutable.

```
theorem models_openDecl_empty : models (Sat (V := V)) (openDecl (∅ : Set N)) = univ
theorem conservative_iff (point : V) (ns : Set N) (d : N) :
    (∀ φ : Sentence N, Avoids d φ →
        Entails (Sat (V := V)) (openDecl (insert d ns)) φ →
          Entails (Sat (V := V)) (openDecl ns) φ) ↔ ns.Nonempty
theorem open_mem_undetermined {ns : Set N} {d : N} (fresh : d ∉ ns) {a b : V} (apart : a ≠ b) :
    Consistent (Sat (V := V)) (insert (.mem d) (openDecl ns ∪ distinctFrom d ns)) ∧
      Consistent (Sat (V := V)) (insert (.nonMem d) (openDecl ns ∪ distinctFrom d ns))
theorem closed_refutes_mem {ns : Set N} {d : N} :
    Entails (Sat (V := V)) (closedDecl ns ∪ distinctFrom d ns) (.nonMem d)
```

The open declaration is the non-perception law once more: before closing, membership of a new name is in doubt; closing makes the declaration perceptible.

8.3 Empty functions

Sets and types answer one classic question differently. Functions out of the empty set into any two codomains form one and the same set. As types, the function types out of the empty type are only equivalent, one for each codomain.

```
theorem funs_empty_eq (y y' : ZFSet.{u}) : ZFSet.funs ∅ y = ZFSet.funs ∅ y'
def emptyFunctionsEquiv (α β : Sort*) : (Empty → α) ≃ (Empty → β)
```

8.4 Parameterized nulls

A family of null values indexed by their kind can be handled by kind or uniformly. The uniform view keeps the kind, re-raising an uncaught null keeps its kind and reason, and a null at every proposition is inconsistent, which is why such nulls may live only in computational types and never in propositions.

```
theorem null_forget_kind {k : Kind} (n : Null k) :
    (Lifted.null n.forget : Lifted T).kindOf? = some k
theorem tryCatch_reraise_kind (n : AnyNull) (clauses : List (Clause T))
    (uncaught : ∀ clause ∈ clauses, clause.catches n = false) :
    (tryCatch (.null n) clauses).kindOf? = some n.1
theorem null_at_every_proposition_inconsistent (nullAt : ∀ P : Prop, P) : False
```

9 What this offers reasoning systems

The techniques the doctrines unify are known, and the honest claim is not that they are new. Local closed-world reasoning, coverage scoring of candidate rules and paraconsistency each exist on their own. What the Buddhist presentation adds is one law with an exact side condition, and with it a

way to state these techniques as instances of one principle instead of as separate rules of thumb. Concretely:

- **Evidence chaining.** A chainer over PLN evidence may convert absence of support into negative evidence only where the source is certified complete for the proposition (Section 4). Blanket closed-world rules, which treat "nothing found" as "evidence against", miscalibrate strength wherever the source is incomplete.
- **Rule learning.** A candidate rule's position on the wheel is its coverage evidence (Figure 3). A *too narrow* rule carries no evidence; negative examples must be certified exclusions, not mere non-memberships.
- **Saturation-based proving.** "Not derived, therefore not entailed" is valid only after a complete saturation; an interrupted run is doubt, never refusal.
- **Conflicting knowledge.** First Degree Entailment over evidence lets a chainer carry contradictory support, the *both* corner, without explosion.

A demonstration would run a PLN chainer whose closed-world steps carry completeness certificates, score rule candidates on the wheel, and keep *both* as a value, measuring calibration on knowledge bases with known gaps against blanket negation as failure.

10 Conclusion

The tradition's central move is to make the condition under which a negative inference holds explicit: perceptibility for non-perception, absolute absence for the third mark, exclusion rather than resemblance for concepts, conventions as commitments for truth. Each condition becomes a hypothesis of a theorem, and each theorem comes with a control showing what goes wrong without it. The result is less a new logic than a gain in semantic clarity: several familiar techniques become instances of a few laws, and the laws state exactly when they apply.

Readings of the texts are interpretations; the theorems are about the formal readings.

A Modules and axioms

Every declaration quoted in this paper is proved in Lean 4 in the Mettapedia library, in the modules below. An audit of all eighty declarations quoted or named in the text finds no `sorry`. Twenty-two depend on no axiom at all; the others depend on at most Lean's three standard axioms, `propext`, `Classical.choice` and `Quot.sound`.

Section	Module
2, 3, 5, 7	`Mettapedia.PLN.RuleFamilies.QuantaleSemantics.DoubleNegationCorners`
	`Mettapedia.PLN.RuleFamilies.QuantaleSemantics.Tetralemma`
3	`Mettapedia.PLN.RuleFamilies.QuantaleSemantics.TwistNegation`
	`Mettapedia.GSLT.Logic.KripkeFixedPoint`
4	`Mettapedia.Logic.NonPerception`
	`Mettapedia.Logic.NonPerception.Instances`
5	`Mettapedia.PLN.RuleFamilies.QuantaleSemantics.ReasonValidity`
6	`Mettapedia.Cybernetics.DistinctionCalculus.Exclusion`
	`Mettapedia.Cybernetics.DistinctionCalculus.ExclusionLearning`
	`Mettapedia.Cybernetics.DistinctionCalculus.ConstructionMiddle`
7	`Mettapedia.Logic.TheoryModel.TwoTruths`
8	`Mettapedia.Languages.MeTTa.Experimental.NoValueZoo`
	`Mettapedia.Logic.TheoryModel.OpenDeclarations`
	`Mettapedia.SetTheory.ZFSet.EmptyFunctions`
	`Mettapedia.Languages.MeTTa.Experimental.NullDesigns`

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